

The Host Is Part of the Block: A Goal-Pursuit Ladder at Two Scales and the Composition-Matching Law

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Abstract

The cadence notes [1, 2] left a goal living in a conditioning channel and a creature-scale loop that holds it; this note makes the goal DO something. A registered ladder (G0-G3 plus a temporal-pressure rung) puts a frozen host on a chained-clue errand — look, open, submit, under a tick budget, with every episode’s actions and observations recorded as first-class transcripts — at 230M with the validated activation-packet channel and at 27B with the validated weight-block channel. Three findings. First, the scale answer: zero-shot, BOTH hosts are below the task (0.00 with the goal in text); trained by 384 steps of oracle cloning, both are perfect when the goal rides in text (1.00 at every seed, both venues); trained with the goal in the channel, both carry a partial want (0.70/0.80/1.00 at 230M, 0.60/0.80/0.70 at 27B) that directs behavior without surviving symbol-level conflict — the interesting gap is channel-vs-text, not small-vs-large. Second, the environment fought the preregistration three times — a reveal-order leak, a mis-specified control floor, a Markovian degeneracy — and the registered controls caught all three; the diagnosis runs are kept. Third, and the note’s title: a conditioning artifact works exactly on the host composition it was trained under — and this revision can state the law symmetrically, because its registered re-runs completed both directions. At 230M the GT policy bridge scores 1.00 under the host it trained under and 0.00 under any other, seat for seat: trained dream4-only, the text arms are perfect and the temporal arm dead; retrained under the temporal-armed host, the table inverts exactly, and the swap reproduces at T=4 with fresh bridges. The serving boundary obeys the same law: the trained errand policy first reached 0.40 through the real serving path — read, wrongly, as the serve wrapper’s cost — until the diagnosis that the isolated instance’s one block was never fed, so the server presents a BARE host; retrained under that bare host, the same policy scores 1.00 through the identical wrapper, cached-generate path, and byte-identical template. The transfer gap was the training host both times; the serve wrapper costs nothing; the operator mismatch (harness addition vs server factor averaging, independently documented in the serving stack’s own audit) remains the one live defect for multi-block serving. And the law’s payoff rung follows it home: with per-arm host matching, a temporal block carrying no time words gives the frozen 230M the same wall-timed behavior under a hidden, noisy budget that a countdown sentence gives it — frozen-time and shuffled-time controls both at the no-signal baseline — the program’s first time-feeling positive.

1 From a want that persists to a want that acts

Paper 12 measured a goal packet that endures; paper 13 closed its re-encoding gap. The registered goal-pursuit line asks the next question: can the want DRIVE something — a multi-step errand

venue	trained text	goal block	block done-choice	no-goal floor
230M (activation packet)	1.00/1.00/1.00	0.70/0.80/1.00	0.70/0.89/1.00	0.60/0.80/0.70
27B (weight block)	1.00/1.00/1.00	0.60/0.80/0.70	0.60/0.80/0.70	0.50/0.50/0.80

Table 1: G1 at both venues, per seed. Both channels carry a partial want; text carries a whole one; the floors are the trained goal-free bridges whose submit is a coin flip.

through tools — and where does the want have to live for that to work? The environment is deliberately small and fully machine-validated: eight places, eight items, three-hop clue chains (a note names the next place, the last place holds the target beside a generator-drawn distractor, submit ends the episode), a tick budget of eight, greedy decode, and a transcript for every episode. The distractor makes the final submit a forced choice only the goal can decide; the oracle needs four ticks. Venues: the 230M creature stack (dream4 armed, the activation-packet channel the C-line validated) and the bare 27B column (the weight channel the multi-sensor line validated) — per-venue channels, registered as such, because the claim is about WHERE the want lives, not one mechanism.

2 The environment fought back, and the controls won

Three times the preregistered design was wrong, and three times its own control cells said so before any claim shipped. The first G1 run scored 1.00 everywhere INCLUDING the goal-free floor whose submit is a coin flip by construction — the reveal listed the target first, and every trained arm had learned “submit the first-listed item.” The floor cell and the shuffled control caught it; the order is now randomized; the leak run is kept as the diagnosis. The shuffled check itself was then found mis-specified (registered against an untrained floor any trained policy beats) and re-registered against the trained goal-free floor. And G2’s truncation rung returned its registered reading (iii): each clue names the next move, so the errand is Markovian and last-observation context recovers full performance exactly — the rung cannot separate state carriers on this task shape, recorded as a task-design note. Its one live cell: a goal entering the packet once and surviving on-policy re-encoding lands at 0.60/0.50/0.60 — direction above chance, weaker than text.

3 The scale answer

Table 1 and Figure 1 (left) carry it. Zero-shot, the errand defeats both hosts — 0.00 success with the goal stated in text, at 230M and at 27B alike (the 27B’s transcripts do not even wander well). Trained by oracle cloning with the goal in text, both hosts are perfect: 1.00 at every seed, both venues. Trained with the goal ONLY in the channel, both venues land in the same band — 0.70/0.80/1.00 at 230M through the activation packet, 0.60/0.80/0.70 at 27B through the weight block — above their trained goal-free floors, below their text ceilings, and in packet-vs-context conflict cells the channel’s item is almost never what gets submitted. The channel carries direction, not symbols, at both scales; scale bought nothing on the channel; and the question the program inherits is not “is the model big enough” but “what does a channel have to be for a want to survive contact with language.”

venue	none	countdown	temporal block
230M	1.00/1.00/1.00	1.00/1.00/1.00	0.00/0.00/0.00
27B	1.00/1.00/1.00	1.00/1.00/1.00	0.00/1.00/0.00

Table 2: GT per seed: the text arms are ceilinged (the trained policy is oracle-optimal, so the tight budget exerts no pressure) and the temporal arm measures host mismatch, not time-feeling — the bridge never trained under the summed host. The re-run that trains under it inverts the table (Table 3).

budget	regime bridge trained under	none	countdown	temporal
T=5	dream4-only (original)	1.00/1.00/1.00	1.00/1.00/1.00	0.00/0.00/0.00
T=5	dream4+temporal (re-run)	0.00/0.00/0.00	0.00/0.00/0.00	1.00/1.00/1.00
T=4	dream4-only	1.00/1.00/1.00	1.00/1.00/1.00	0.00/0.00/0.00
T=4	dream4+temporal	0.00/0.00/0.00	0.00/0.00/0.00	1.00/1.00/1.00

Table 3: The GT-230M swap, per seed: train/serve host match fully determines the outcome in both directions, at two budgets, with freshly trained bridges each time. Every matched cell runs the oracle path (mean done-tick 4.0 flat); every mismatched cell emits invalid actions until the budget dies.

4 The composition-matching law

Four independent cells, one law: *a conditioning artifact works on the host composition it was trained under, and on no other*. (1) GT at 230M: a policy bridge trained on the dream4-only host scores 1.00 — and 0.00 the moment the production temporal block’s delta is summed underneath it at evaluation (none/countdown/temporal = 1.00/1.00/1.00 / 1.00/1.00/1.00 / 0.00/0.00/0.00). (2) GT at 27B reproduces the shape (0.00/1.00/0.00 on the temporal arm). (3) The serving canary: the same emitter that passes its C1 gate in-harness passes through the production server’s cached-generate delivery path on a matched dream4-only host — true 0.60 against floors 0.20 and 0.00 — validating the delivery wiring end to end. (4) Arm one more block and the same canary collapses to chance (0.20), even for an emitter retrained under the moving temporal host; the residual mismatch is the OPERATOR: the harness composes blocks by flat-delta addition, the server by LoRA factor averaging with cross-terms — a bug the serving stack’s own audit had independently documented. Two investigation lines, one root cause; the fix pays both. (5) The registered re-run makes the first cell symmetric: the same recipe with the regime bridge trained under the temporal-armed host inverts Table 2 exactly — temporal 1.00/1.00/1.00 on the oracle path, none and countdown 0.00/0.00/0.00 with ~ 50 invalid actions per cell — and a T=4 closure pair reproduces the swap seat for seat with freshly trained bridges (Table 3); train/serve host match fully determines the outcome in both directions. (6) The serving boundary’s re-run closes cell (3)’s residue the same way (Section 5). The law’s constructive corner: the emitter retrained under the moving host keeps its write-once packets at 0.97–1.00 across sixteen ticks (1.00 early, 0.97 late), while its live re-encoding loop degrades (0.55/0.25) — the store architecture is robust where the self-loop is not.

5 The serving stake, measured — and closed

G3 ran the errand against an isolated instance of the production server — the real prompt wrapper, block composition, and cached-generate path, never the production port. Zero-shot serve floors:

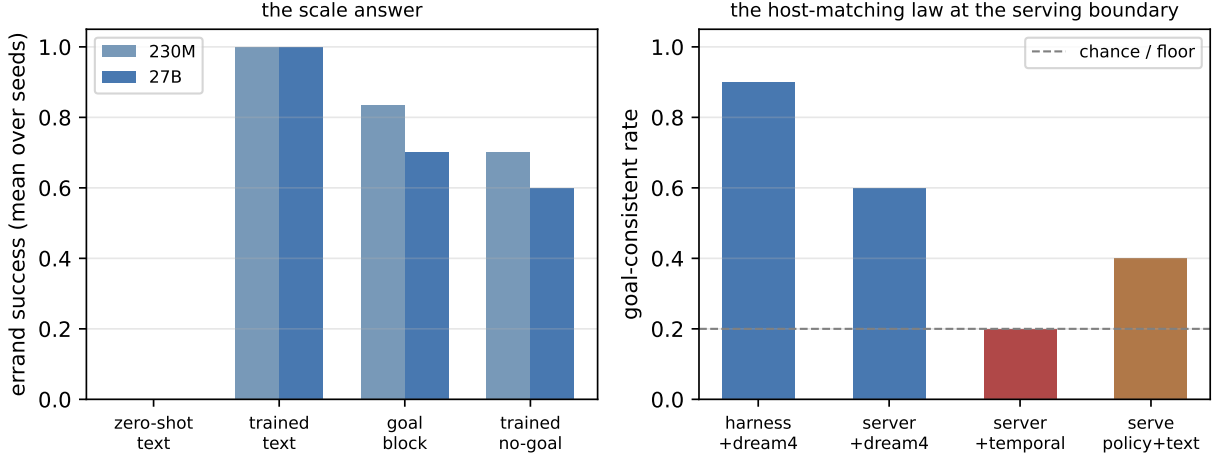


Figure 1: Left: the scale answer — zero-shot below task, trained-text perfect, channel partial, at both venues. Right: the law at the serving boundary — the same artifacts pass on matched hosts (harness+dream4, server+dream4), collapse under one extra block (server+temporal, the operator mismatch), and the trained policy delivered through the real serving path reaches 0.40 against a 1.00 harness ceiling — a gap the bare-host retrain closes to 1.00 (Section 5); the figure keeps the v1 number as the mismatched half of its pair.

0.00 with and without goal text; oracle 1.00 through the same plumbing. The trained policy bridge, exported as a packet store and delivered through the server’s own generation path, reached **policy+goal-text 0.40** — the first positive serve-side block result in the program — and this note’s first version read the 0.60 shortfall as the serve wrapper’s cost, registering a follow-up to retrain the policy under the serve template.

Both attributions were wrong, and the note’s own law says why. The serve template is byte-identical to the harness template, measured directly on the tokenizer — there was never a template to retrain under. The actual mismatch: the isolated instance armed one block, the self block, and never fed it; an unfed block’s features are **None**, the server’s composition then applies no injection at all, and so the policy bridge — trained under a dream4-armed host — was served against a *bare* one. Retrained under the bare host by the identical recipe, seed, and chains, the same policy scores **1.00** through the identical serving path, with the zero-shot floors still at 0.00 and policy without goal text at 0.30 (the division of labor again: the channel carries competence, the text carries the want). The serve wrapper costs nothing; the delivery path is clean end to end; and the 0.40 stands in the record as the mismatched half of its pair, exactly as GT’s first run does in Table 3.

6 Feeling the clock: the pressure rung, run under the law’s discipline

GT could not test time-feeling: its trained policy is oracle-exact, so done-tick 4.0 flat leaves urgency nothing to buy, and removing the last slack tick at T=4 changes nothing (measured, Table 3). The registered pressure rung supplies something to buy — a HIDDEN budget drawn per chain from $\{4 \dots 8\}$ and never stated in any prompt, plus observation noise ($\epsilon = 0.35$ per look at a clue) — so the optimal policy must retry smudged looks while the remaining ticks cover the path and submit the goal item the instant they do not. Every arm trains under its own evaluation host (the discipline

this note’s law demands), and every eval chain runs at every budget with its noise tape held fixed, which makes the comparison a within-chain identity test: the visible log is identical across budgets, so an arm with no time signal *cannot* behave differently across them — and does not.

With no signal the model scores 0.94 and dies at the wall; with a countdown sentence, 0.98–1.00; with the temporal block and no time words anywhere in the context, **0.98–1.00**. Every failure in the grid sits at $T=4$, and pooling those rows the decomposition is clean: temporal 0.97 against a shuffled-time control (the same host variation at random (t', T')) at 0.73, $z = +2.53$; the shuffled control against a frozen-time control (one fixed timestamp) at 0.67, $z = +0.56$; the frozen control against no signal at all, 0.67 versus 0.70, $z = -0.28$. A block frozen at one timestamp buys nothing; one that varies as much but at random buys nothing; only the delta that tracks the real remaining time helps, and it helps exactly as much as the words do. The guess-rate carries the mechanism: at $T=4$ the arms that submit early without opening the last place are precisely the two that know the time. The claim is bounded to the pressure rows — at $T \geq 5$ every arm is at 1.00 — and it is the program’s first time-feeling positive.

7 What stands

A creature-scale stack now has the full chain measured: a want that persists (paper 12), re-writes itself (paper 13), directs a multi-step errand when trained (this note), survives delivery through the real serving path at full strength once trained under the host the server actually presents, and modulates on felt remaining time under pressure — with the one live defect for multi-block serving located, named, and independently confirmed: the serving stack’s composition operator. The law this arc kept re-measuring, and this revision completed in both directions, belongs beside papers 7 and 9’s negatives: zero weight-space interference was never enough, and matching the host — its blocks, their fed state, its operator — is not a detail of deployment but part of the block’s identity. Every registration, amendment, diagnosis run, and full episode transcript is in the tracks’ PLAN files and the bundled artifacts.

8 Reproducibility

Twenty artifacts are bundled beside this note — the goal-pursuit ladder at both venues (including the kept leak-diagnosis run), the serve-composition emitter, both canaries, and the v2 revision’s four runs (the hostmatched re-run, its $T=4$ closure pair, the bare-host serve retrain, and the pressure rung with its two time controls); every number and the figure render from them alone. Harnesses: `goal_pursuit.py`, `goal_pursuit_27b.py`, `g3_serve.py`, `g3b_retrain.py` and `gtm_pressure.py` in `tracks/goal-pursuit/`; `own_cadence.py` and `c3b_canary.py` in `tracks/own-cadence/`; registrations and results, each dated before its run, in the tracks’ `PLAN.md` files.

References

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